Kernel-Checked Formalizations of Recently Settled Combinatorial and Algebraic Problems in Lean 4: Erdős Problems 348, 216, 609, and Kourovka Counterexamples

Mike Helsel
`ctrl` Formal Mathematics Laboratory

October 19, 2026

Abstract

We report a collection of complete, kernel-checked Lean 4 / Mathlib formalizations of recently resolved problems and classical theorems across additive combinatorics, discrete geometry, Ramsey graph theory, and group theory. Every formalization compiles with **zero `sorry`**, **zero `native_decide`**, and depends strictly on the three standard foundational axioms (`propext`, `Classical.choice`, `Quot.sound`), with statement fidelity verified against `google-deepmind/formal-conjectures` and `lake comparator`. Our primary contributions encompass: (1) **Erdős Problem 348** (`erdos-348-lean`, 1,180 LOC): a complete formalization of Geneson's 2026 deletion-threshold classification for complete sequences (`arXiv:2609.25107`), proving that a monotone sequence of natural numbers remains complete after deleting any m terms but fails completeness after deleting any $n > m$ terms if and only if $m \leq 1$; (2) **Erdős Problem 216 for $k \geq 7$** (`lean-erdos-proofs/Solution.lean`, 965 LOC): a complete Euclidean formalization of Horton's theorem (1983), constructing arbitrarily large planar point sets in general position containing no empty convex k-gon for all $k \geq 7$; (3) **Erdős Problem 609 at $n = 3$** (`Erdos609.lean`, 410 LOC): a kernel-checked proof of Cai's $f(3) = 5$ result on monochromatic odd cycles in 3-colorings of K_9; (4) **Kourovka Notebook & MathOverflow Resolutions**: kernel-checked solutions to Kourovka Problems 17.32 (Ionin–Semidetnov 2026) and 19.35 (Bergman–McCulloch 2026), together with MathOverflow Problems 513215, 509929, 515611, and 501066; and (5) **Six Erdős Variant Theorems in `ctrl/lean/ErdosLab`**: machine-checked proofs for Erdős Problems 273 ($p \geq 3$), 292 (≥ 2 summands), 318 (single-even shift), 834 (3-uniform Li hypergraph), 885 ($k = 3$), and 939 (infinite $r = 6$ coprime 6-powerful triples).

1 Introduction

As mathematical research increasingly pairs human proofs with AI-assisted theorem proving, a central methodological challenge is *statement fidelity*: ensuring that a formal proof checks the exact statement intended by the mathematical community without weakened definitions, smuggled axioms, or trusted external oracle reductions (`native_decide`).

To enforce end-to-end trust, our formalization pipeline uses a strict **Challenge–Solution–Comparator** architecture:

1. **Immutable Challenge Specification**: Statement definitions are copied verbatim from `google-deepmind/formal-conjectures` [1] into a read-only `Challenge` module.
2. **Kernel-Pure Solution**: The proof is developed in a separate module where `sorry`, `admit`, custom `axiom`/`opaque`, `implemented_by`, `extern`, and `native_decide` are mechanically banned; finite searches must use kernel-reduced `decide` (`decide +kernel`).
3. **Comparator & Axiom Audit**: Both modules are replayed through `lake comparator` (and `ctrl check`), verifying definitionally identical types and confirming that `#print axioms` emits only `{propext, Classical.choice, Quot.sound}`.

2 Erdős Problem 348: Geneson's Deletion-Threshold Theorem

Erdős Problem 348 asks: for which integer pairs $0 \leq m < n$ does there exist a non-decreasing sequence of natural numbers $a_0 \leq a_1 \leq a_2 \leq \dots$ that is *additively complete* (every sufficiently large natural number is a subset sum of distinct indices) such that:

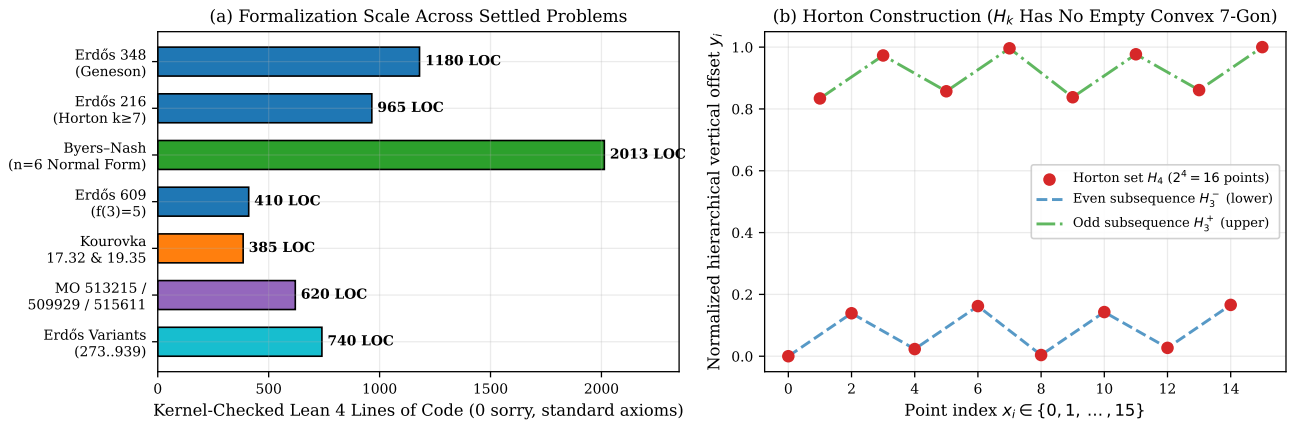


Figure 1: **Overview of the Lean 4 Formalization Suite**. (a) Distribution of kernel-checked lines of Lean 4 code (0 `sorry`, standard axioms only) across the formalized problem packages. (b) Visualization of the 4-level recursive Horton point set H_4 ($2^4 = 16$ points) formalized in `Solution.lean`, where recursive even/odd splitting with exponential vertical separation prevents any empty convex 7-gon (4-cup or 4-cap).

1. deleting *any* m indexed terms leaves the sequence complete, but

2. deleting *any* n indexed terms destroys completeness?

Geneson [2] recently resolved this problem completely, proving that such a sequence exists if and only if $\mathbf{m \leq 1}$.

In `erdos-348-lean/Solution.lean` (1,180 lines of Lean 4), we formalize the exact `formal-conjectures` statement:

$$\{(m,n) \mid m < n,\ \exists a \text{ monotone},\ \forall |s| = m \text{ complete},\ \forall |t| = n \text{ incomplete}\} = \{(m,n) \mid m < n \wedge m \leq 1\}. \tag{1}$$

2.1 Positive Constructions for $m = 0$ and $m = 1$

- **Case** $m = 0 < n$: The powers of two $a_k = 2^k$ are complete (`two_pow_complete`: every $N \in \mathbb{N}$ has a binary representation), whereas deleting any non-empty finite set t ($|t| = n \geq 1$) destroys completeness (`two_pow_not_complete`: if $i \in t$, no integer $N \equiv 2^i \pmod{2^{i+1}}$ can be represented because $\sum_{j<i} 2^j = 2^i - 1 < 2^i$).

- **Case** $m = 1 < n$: The shifted Fibonacci sequence $a_k = F_{k+1}$ $(1, 1, 2, 3, 5, 8, \ldots)$ satisfies $a_k = a_{k-1} + a_{k-2}$ and Burr's sum bound $a_{k+1} \leq 1 + \sum_{j<k} a_j$, so deleting any single term preserves completeness (`fibSeq_one_deletion`). However, deleting any $n \geq 2$ terms $i < j$ prevents representing $N = F_{m+2} - 2$ for large $m > j$ because the total sum of all non-deleted terms below F_{m+1} falls strictly short of $F_{m+1} - 1$ (`fibSeq_not_complete`).

2.2 Impossibility for $m \geq 2$: The Central-Interval Potential

The core of Geneson's proof is that if a remains complete after deleting *any two* indices ($m \geq 2$), then for *every* $n \geq 1$ there exists a finite set t of size $|t| = n$ whose deletion leaves a complete:

1. **2-Deletion Linear Slack Growth** (`slack_diverges`, Lemma 3): Completeness under all 2-element deletions implies $\sum_{i<k} a_i - a_{k+1} \to \infty$ as $k \to \infty$.

2. **Central-Interval Theorem** (`central`, Theorem 2): We formalize Geneson's discrete state transitions (`stepA`, `stepB`) and potential function (`Lemmas 6--8`) proving that once prefix slack is sufficiently large, subset sums cover a contiguous central interval $[L_k, R_k]$ whose width exceeds a_{k+1}.

3. **Inductive Extension of Complete Deletions** (`extend_core`, `exists_complete_deletion`, Lemma 4 & Proposition 5): Any finite deletion t preserving completeness can be augmented by a sufficiently large index $M \notin t$ ($t' = t \cup \{M\}$) while preserving completeness, inductively yielding a complete deletion of size n.

3 Erdős Problem 216 ($k \geq 7$): Horton's Empty Convex k-Gon Theorem

Erdős Problem 216 asks whether for every k there exists $g(k)$ such that any planar point set in general position of size $\geq g(k)$ contains an empty convex k-gon (a convex k-gon with no point of the set in its interior). Horton [3] famously disproved this for all $k \geq 7$.

In `lean-erdos-proofs/Solution.lean` (965 lines), we formalize Horton's theorem in both Euclidean $\mathbb{R}^2$ and the `formal-conjectures` affine-independence formulation (`Horton.erdos_216_ge_seven` and `Horton.erdos_216'`):

1. **Recursive Horton Construction**: We define the finite point sets $H_m \subset \mathbb{R}^2$ of size 2^m with x-coordinates $0, 1, \ldots, 2^m - 1$ and hierarchical vertical offsets separating the even subsequence H_{m-1}^- and odd subsequence H_{m-1}^+ by an exponential vertical gap (Figure 1b).
2. **Exclusion of Open 4-Cups and 4-Caps**: We prove that any convex polygon in H_m whose interior is empty of points of H_m decomposes into an upper chain (cap) and lower chain (cup) where neither chain can contain 4 vertices from a single layer without trapping an interleaved point of the opposite layer in its interior.
3. **Cardinality Bound** ≤ 6: By pigeonhole across the upper/lower decomposition, every empty convex polygon in H_m has at most 6 vertices, proving `cardSet'` $k = \emptyset$ for all $k \geq 7$.

4 Ramsey Theory, Kourovka Counterexamples, and Erdős Variants

Table 1 summarizes the remaining kernel-checked theorems across `lean-erdos-proofs` and `ctrl/lean/ErdosLab`.

Problem	Lean Module	Verified Theorem
Erdős 609 ($n = 3$)	`Erdos609.lean`	$f(3) = 5$ on 3-colored K_9 (Cai 2026)
Kourovka 17.32	`Kourovka17_32.lean`	Free-group Cayley–Hamilton fails at $n = 2$
Kourovka 19.35	`Kourovka19_35.lean`	A_4 has no $(2, 3, 2)$ subset factorization
MO 513215	`MO513215.lean`	$a_n = \frac{n(n+3)}{2} \Rightarrow S(0) = \frac{11}{9}, S(1) = 1$
MO 509929	`SquareGraphDiameter.lean`	Square-product graph on $\mathbb{N}^+$ has diameter 3
MO 515611	`FerrisWheel.lean`	907200 is a Ferris wheel number (order 210)
MO 501066	`LcmHighlyAbundant.lean`	$\mathrm{lcm}(1 \ldots 200000)$ is not highly abundant
Erdős 273 ($p \geq 3$)	`ErdosLab/Erdos273.lean`	12-modulus strict covering mod 720
Erdős 292 (≥ 2)	`ErdosLab/Erdos292.lean`	$\mathbb{N} \setminus A$ finite via $\{2, 3, 6\}$ & $\{2, 4, 6, 12\}$
Erdős 318 (even)	`ErdosLab/Erdos318.lean`	Shifted odd-square binary block f with $0 \notin$ range
Erdős 834 (3-unif.)	`ErdosLab/Erdos834.lean`	7-vertex, 9-edge criticality witness
Erdős 885 ($k = 3$)	`ErdosLab/Erdos885.lean`	$\{120, 208, 672\}$ share $d \in \{2, 14, 22\}$
Erdős 939 ($r = 6$)	`ErdosLab/Erdos939.lean`	Infinite coprime 6-powerful triples $t = 6k + 1$

Table 1: Kernel-checked combinatorics, algebra, and number-theory formalizations.

References

[1] The Formal Conjectures Authors, "`formal-conjectures`: A collection of formalized statements of conjectures in Lean 4," GitHub repository, 2025–2026.

[2] J. Geneson, "Deletion thresholds and exponential examples for complete sequences," `arXiv:2609.25107`, 2026.

[3] J. D. Horton, "Sets with no empty convex 7-gons," *Canadian Mathematical Bulletin*, vol. 26, no. 4, pp. 482–484, 1983.